



Department of Electronic & Telecommunication Engineering
University of Moratuwa
Sri Lanka.

Analog Audio Effects Processor

Group Name : Team SuperTronics

Group Members :

230045X	Anthony C.S.B
230140J	Dharmakeerthi P.K.G.C.L
230477X	Perera H.A.J.I
230730T	Wijethilaka J.S.

Submitted in partial fulfillment of the requirements for the module
EN 2091 Laboratory Practice and Projects

Date : 14-12-2025

Abstract

This project presents the design and implementation of an analog audio effects processor which combines multiple commonly used audio effect into a single modular platform. The modules present in the project includes a power supply module capable of delivering a $\pm 12\text{V}$ dual rail linear power supply, an input buffer, a three band audio equalizer, three audio effects namely as tremolo, phaser and distortion and an output buffer. The dual rail linear power supply is employed to ensure stable and low noise operation of the analog circuitry.

Each functional module is implemented using operational amplifier based circuits with careful attention given to impedance matching, maintaining signal integrity, noise reduction and maintaining proper gain control. Modulation effects such as tremolo and phaser are realized using low frequency oscillators and optical control elements (optocouplers) to achieve smooth and musically present variations. The distortion stage employs diode based clipping to introduce controlled non linearities suitable to achieve the desired effect.

The system was designed simulated and tested using industry standard software. Experimental results demonstrated that the processor successfully produced the intended audio effects while maintaining stable operation and acceptable signal quality. This project provides practical insight into analog audio circuit design and highlights the effectiveness of modular architecture in multi effect systems.

Abbreviations and Acronyms

AC	Alternating Current
DC	Direct Current
LFO	Low Frequency Oscillator
EQ	Equalizer
PCB	Printed Circuit Board
LED	Light Emitting Diode
LDR	Light Dependent Resistor
IC	Integrated Circuit
RC	Resistor-Capacitor
PSU	Power Supply Unit
VCA	Voltage Controlled Amplifier

Contents

1	Introduction and Functionality	4
2	System Architecture	4
2.1	Table and Figure	5
2.2	Equations	5
2.3	Guidelines for References	6
3	Component Selection	6
4	PCB Design	7
5	Enclosure Design	8
6	Software Simulation and Hardware Testing	9
6.1	LTspice Simulations	9
6.2	Hardware Testing	10
7	Conclusion & Future Works	11
8	Contribution of Group Members	11

1 Introduction and Functionality

Audio Effects play a vital role in audio signal processing by modifying the tonal, temporal and harmonic characteristics of the input audio signal. While modern digital signal processing techniques are widely used, analog audio effects remain popular due to their natural sound quality, continuous time operation and musical response. Designing analog audio effects also provides strong educational value as it requires a sound understanding of operational amplifiers, passive components and signal conditioning techniques.

The objective of this project is to design and implement an analog audio effects processor capable of processing an audio input signal through several commonly used audio effects. The system is intended to demonstrate practical applications of analog electronics concepts such as amplification, filtering, modulation and non-linear signal processing.

The developed system consists of several modular stages each performing a specific function. An input buffer is used to match the impedance between the guitar pickup and the processing circuitry. A three-band equalizer provides frequency shaping through bass, mid and treble controls. Modulation-based effects such as tremolo and phaser are implemented using LFOs and optical control elements. A distortion stage introduces controlled non-linear clipping to produce overdrive effects. Finally, an output buffer ensures stable signal delivery to external audio output equipment.

The overall system accepts audio input through a 3.5mm audio input, processes it through the selected effects while preserving signal integrity and produces a clean, low noise audio output.

2 System Architecture

The audio effects processor is designed using a modular system architecture where each functional block operates independently while remaining electrically compatible with the rest of the system. This approach simplifies design, testing, debugging and future expansion. The system is powered by a centralized dual rail power supply that provides $\pm 12\text{ V}$ to all analog stages.

The signal flow begins at the audio input jack where the incoming audio signal is first passed through an input buffer. This buffer provides high input impedance and prevents signal loading. The buffered signal is then routed through the equalizer stage, allowing tonal adjustment before modulation effects are applied. The processed signal subsequently passes through the tremolo and phaser stages where time-varying modulation effects are introduced. A distortion stage follows, adding harmonic content through diode-based clipping. Finally, an output buffer delivers the processed signal to the output jack with low output impedance.

Careful attention has been given to inter-module interfacing. AC coupling capacitors are used between stages to block DC offsets, while buffer amplifiers prevent impedance mismatches. Local decoupling capacitors are placed near op-amp power pins to minimize noise and ensure stable operation.

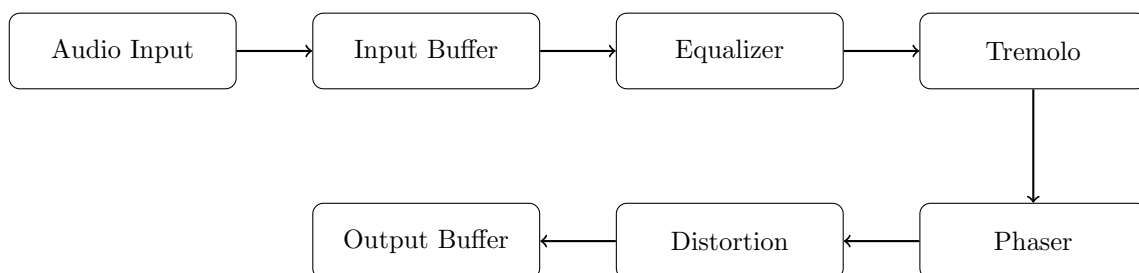


Figure 1: System architecture of the analog multi-effects audio processor

2.1 Table and Figure

Figures and tables are used in this report to clearly illustrate the structure and functionality of the proposed analog audio effects processor. Figure 1 presents the overall block diagram of the system, highlighting the modular signal flow from the audio input stage to the output buffer. The diagram demonstrates the sequential arrangement of the input buffer, equalizer, modulation effects, distortion stage and output buffer.

Table 1 summarizes the main functional modules of the system along with their respective functions and key components used in each module. The table provides a concise overview of the system organization and aids in understanding the role of each block within the complete audio processing chain.

Table 1: Functional modules of the audio effects processor

Module	Function	Key Components
Input Buffer	Impedance matching	TL072 op-amp
Equalizer	Frequency shaping	Op-amps, RC networks
Tremolo	Amplitude modulation	LFO, LED-LDR
Phaser	Phase modulation	LFO, LED-LDR
Distortion	Non-linear clipping	Diodes, op-amp
Output Buffer	Signal driving	TL072 op-amp

2.2 Equations

The operation of the proposed analog audio effects processor can be explained using a set of fundamental equations related to amplification, filtering and modulation. These equations describe the behavior of key circuit blocks implemented in the system.

The voltage gain of the operational amplifier stages used in the input buffer, equalizer, distortion and output buffer modules is determined using the standard non-inverting amplifier gain expression given by

$$A_v = 1 + \frac{R_f}{R_{in}}, \quad (1)$$

where R_f represents the feedback resistance and R_{in} denotes the input resistance.

Frequency-selective behavior in the equalizer and filtering stages is characterized by the RC cut-off frequency, which is given by

$$f_c = \frac{1}{2\pi RC}, \quad (2)$$

where R and C are the resistance and capacitance values of the respective filter network.

The tremolo effect is based on the principle of amplitude modulation, in which the amplitude of the audio signal is varied by a low-frequency control signal. This behavior can be represented as

$$y(t) = x(t) [1 + m \sin(2\pi f_{LFO} t)], \quad (3)$$

where $x(t)$ is the input audio signal, m is the modulation depth, and f_{LFO} is the frequency of the low frequency oscillator.

All equations are used to guide the design and analysis of the corresponding circuit blocks and support the functional behavior of the implemented audio effects.

2.3 Guidelines for References

The reference list is included in the *References* section at the end of this document. Sources are cited within the text using numbered references enclosed in square brackets. Each reference number corresponds to an entry in the reference list and appears in the order of first citation.

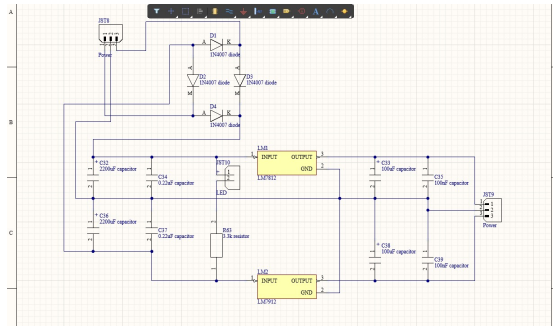
The references used in this project include textbooks on analog electronics and audio circuit design, as well as manufacturer datasheets for the components employed in the system.

3 Component Selection

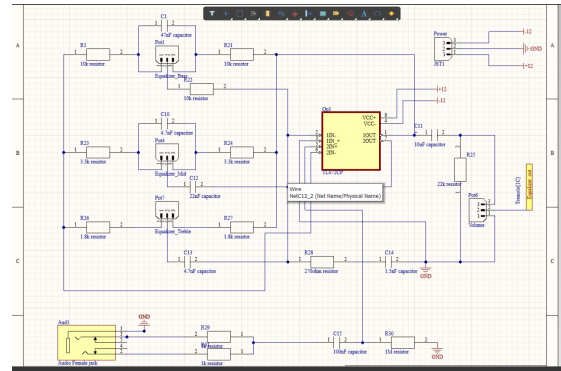
Component selection was carried out with careful consideration of audio performance, reliability, availability, and suitability for analog signal processing. The selected components ensure stable operation, low noise, and effective realization of the intended audio effects.

- **Operational Amplifier (TL072):** The TL072 operational amplifier was chosen for the majority of signal processing stages including the input buffer, equalizer, distortion, tremolo, phaser and output buffer. This device features low input noise, JFET input transistors providing high input impedance and a wide bandwidth with a high slew rate suitable for audio based applications. The dual op-amp configuration also reduces component count and simplifies circuit implementation.
- **Voltage Regulators (LM7812 and LM7912):** The dual-rail power supply was implemented using LM7812 and LM7912 voltage regulators to provide regulated ± 12 V DC outputs. These linear regulators were selected due to their reliability and low output ripple, which are important characteristics for analog audio circuits.
- **Rectification and Filtering Components:** Bridge rectification was achieved using 1N4007 diodes, selected for their high reverse voltage rating and suitability for low frequency rectification. Electrolytic capacitors were used for bulk smoothing of the rectified voltage, while ceramic capacitors were employed for high-frequency decoupling to improve power supply stability.
- **Resistors and Capacitors:** Resistors and capacitors were selected to form gain-setting, biasing, and frequency selective networks throughout the system. RC networks were designed to operate within the audible frequency range and to ensure stable and predictable circuit behavior.
- **Potentiometers:** Potentiometers were incorporated in the equalizer, tremolo, phaser and distortion stages to provide user adjustable control over gain, frequency, modulation rate and modulation depth. These components were selected to allow precise adjustment of circuit parameters ensuring that the required frequency response and gain characteristics could be achieved under varying operating conditions.
- **Optical Components (LEDs and LDRs):** Light-emitting diodes (LEDs) and light-dependent resistors (LDRs) were used in the tremolo and phaser modules to implement optical modulation. This approach helps achieve simple and smooth modulation of the audio signal.
- **Diodes for Distortion (1N4148):** Silicon signal diodes (1N4148) were employed in the distortion stage to introduce controlled non linear clipping. These diodes were selected due to their fast switching characteristics and consistent forward voltage which enable predictable and repeatable distortion behavior suitable for audio applications.

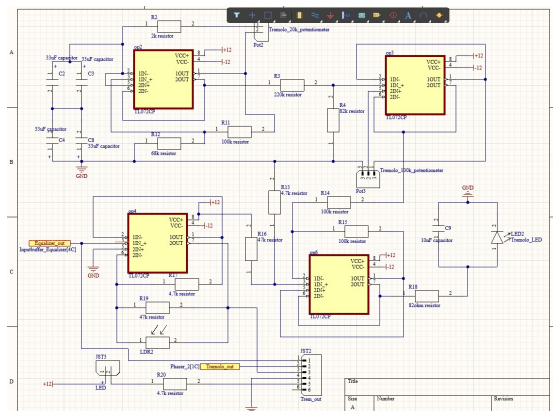
4 PCB Design



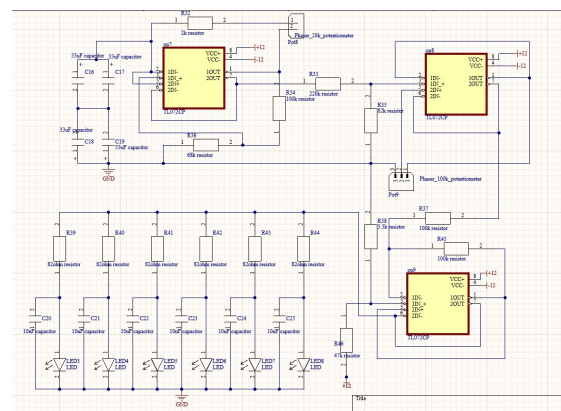
(a) Power Supply schematic



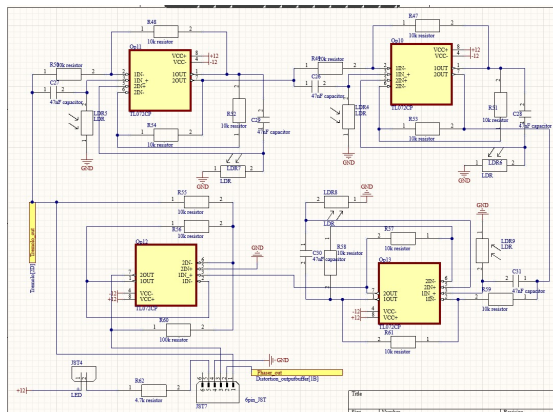
(b) Input buffer and equalizer schematic



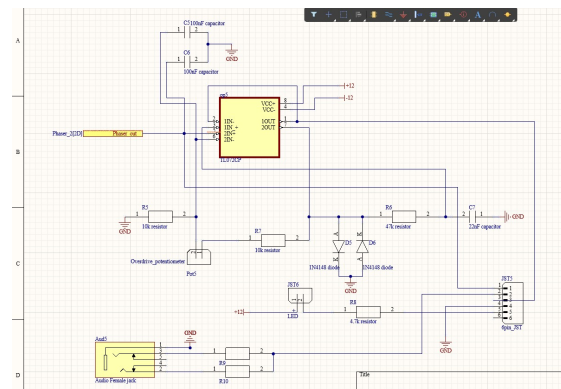
(c) Tremolo schematic



(d) Phaser schematic (1)



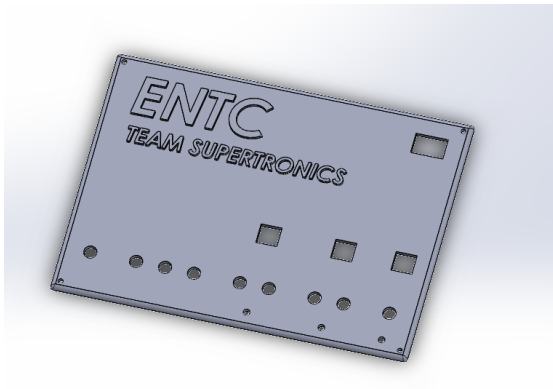
(e) Phaser schematic (2)



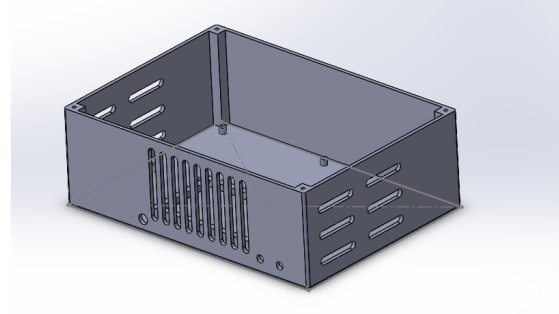
(f) Distortion and output buffer schematic

Figure 2: Schematics of the main functional modules of the analog multi-effects audio processor

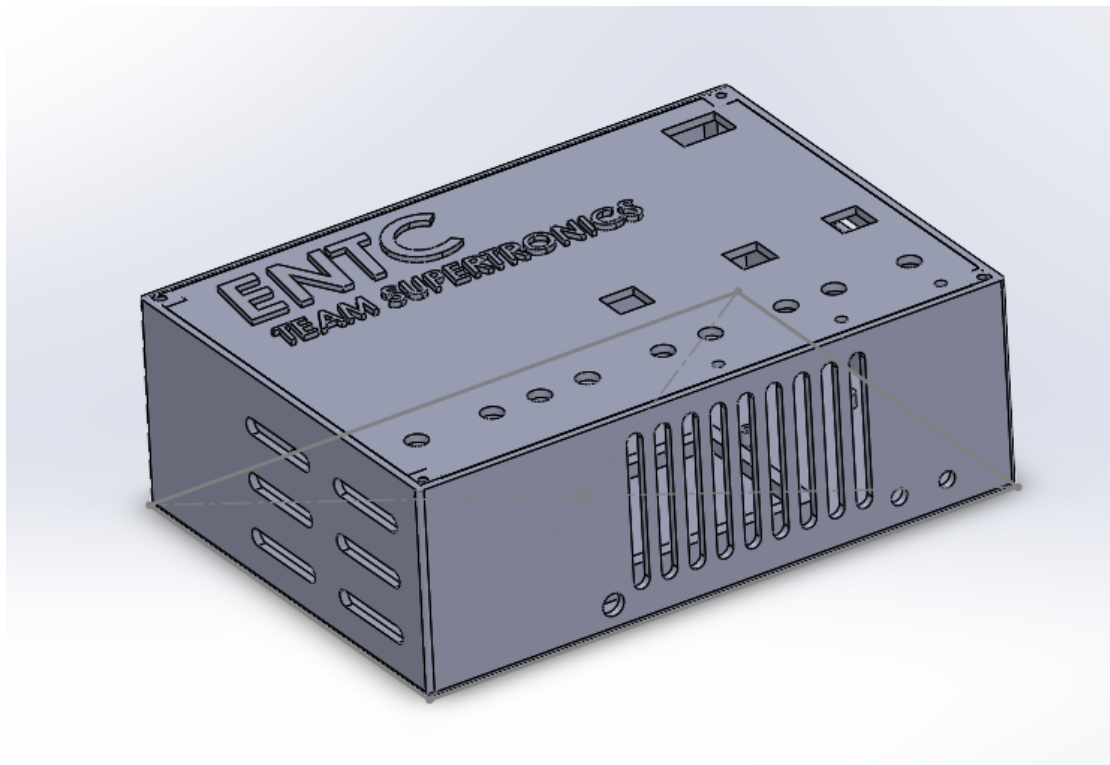
5 Enclosure Design



(a) Upper Part



(b) Lower Part



(c) Enclosure

6 Software Simulation and Hardware Testing

6.1 LTspice Simulations

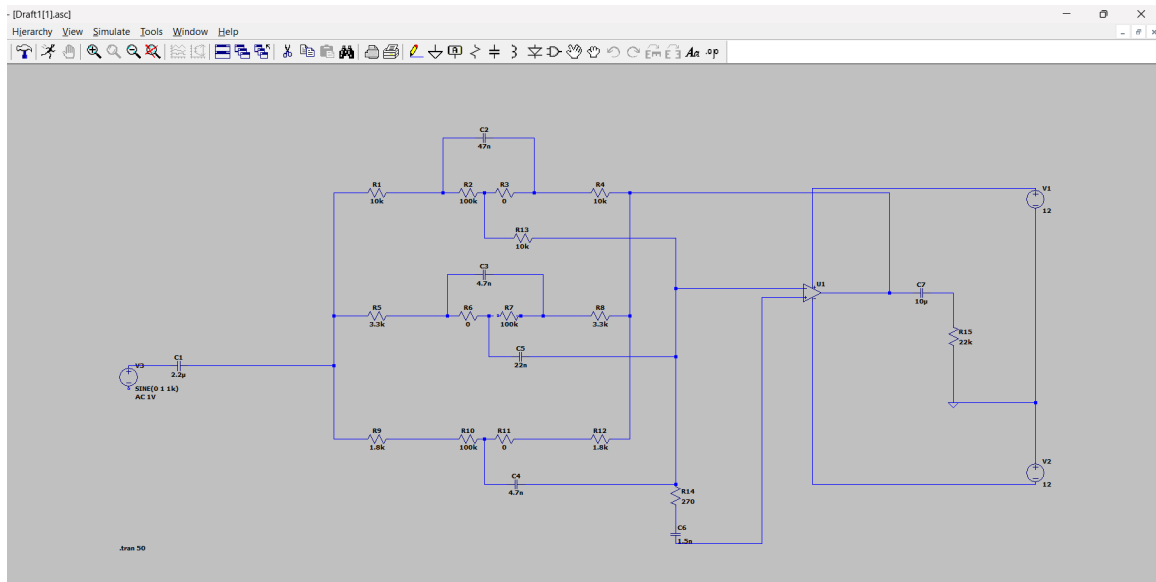


Figure 4: Equalizer

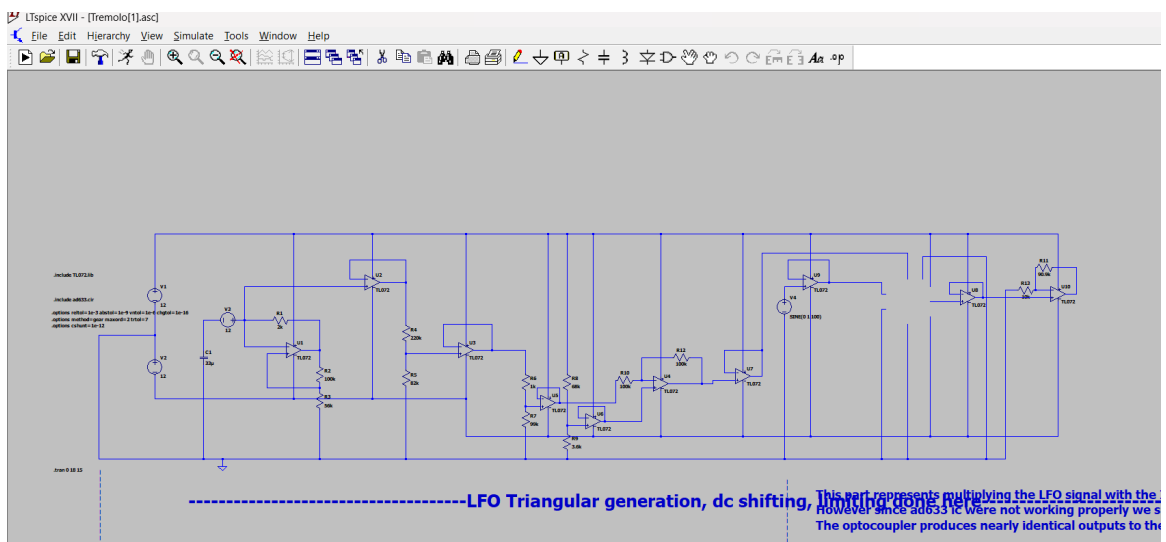


Figure 5: Tremolo

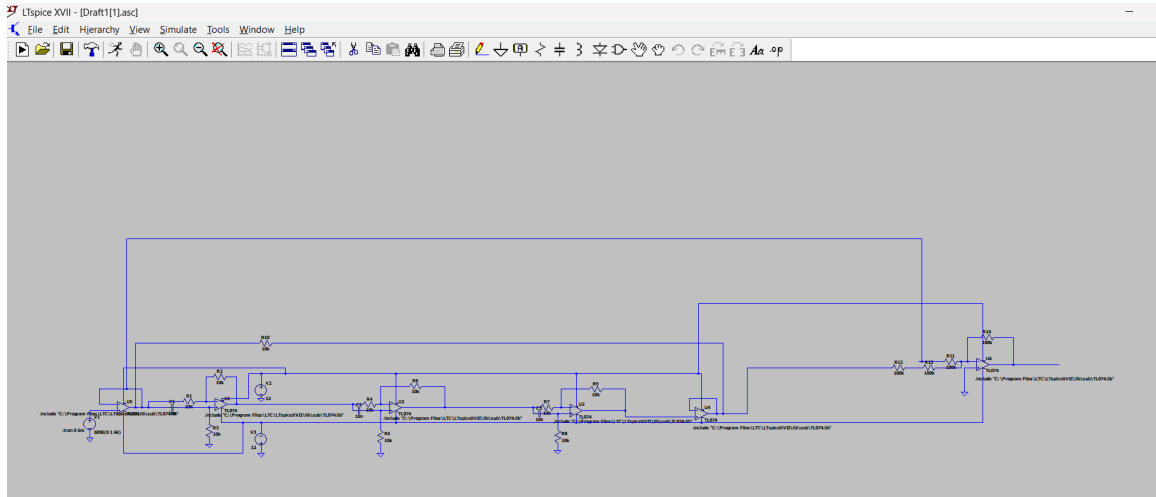


Figure 6: Phaser

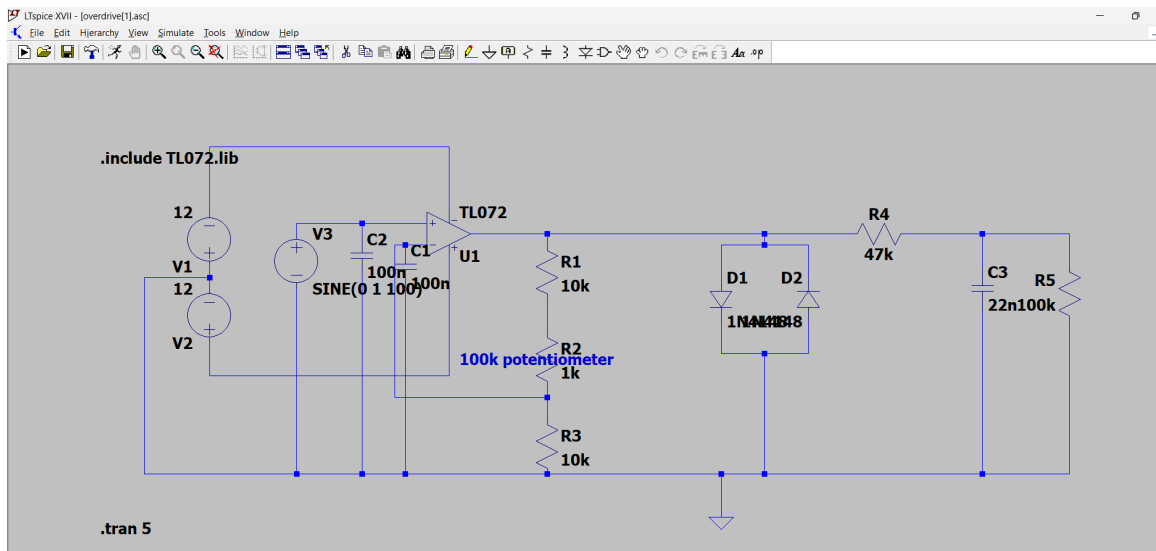


Figure 7: Overdrive

6.2 Hardware Testing

Each module was first simulated in LTSpice to verify its functionality. Subsequently, the modules were implemented on breadboards, and their performance was tested to ensure correctness. The observed outputs on the breadboards matched the simulation results, confirming that the desired behavior was successfully achieved in the hardware implementation.

7 Conclusion & Future Works

We successfully achieved the design and implementation of an analog audio effects processor intended for audio signal processing. The system integrates several commonly used audio effects including a three band equalizer, tremolo, phaser and distortion into a single modular platform powered by a dual-rail ± 12 V supply. The modular architecture adopted in the design facilitated systematic development, testing, and integration of individual functional blocks.

Hardware testing confirmed stable operation of the power supply and reliable performance of all signal processing stages. Each effect module produced the intended audible characteristics while proper buffering and grounding ensured acceptable signal quality with acceptable noise. Through this project we gained valuable practical experience in operational amplifier based circuit design, analog signal processing, PCB layout considerations and system level integration of multiple electronic subsystems.

Several enhancements can be considered as future work to further improve the functionality and performance of the system. Digital control of effect parameters using a microcontroller could be introduced to enable preset storage and improved user interaction. Additional audio effects such as delay, chorus and reverb could be incorporated to expand the versatility of the processor.

8 Contribution of Group Members

Index	Tasks
230045X	Overdrive Module, PCB Design
230140J	Phaser Module, Enclosure Design
230477X	Tremolo Module, Component Selection, Assembling
230730T	Audio Equalizer, Power Supply

Appendix

References

- [1] B. Cordell, *Designing Audio Circuits and Systems*, Taylor & Francis Ltd, 2024.
- [2] J. C. M. Lam, *Analog Audio Amplifier Design*, Focal Press, 2024.